OTHER GLMS

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THREE SO FAR

- Poisson
- Binomial
- What's the third?
- Gaussian (aka linear regression)
- Includes all members of the exponential family
- Same code can be used - just change the `family` argument

GAMMA

Waiting time distribution
Data are:
- Continuous
- Greater than zero

Some details
$y_i \sim \Gamma(\alpha, \beta)$
- Inverse link function

INVERSE GAUSSIAN

Time to event data, frequencies
Data are:
- Continuous
- Greater than zero

Some details:
$y_i \sim IGaus(\mu, \lambda)$
where, $\lambda = \frac{\mu^3}{\sigma^2}$

- Inverse square link function

BETA

Probabilities
Data are:
- Continuous
- Between 0 and 1

Some details:
$y_i \sim \beta(\alpha, \beta)$

- Logit link
- Not a GLM mathematically, but a GLM practically

MULTINOMIAL

Data fall in multiple categories
Data are:
- Discrete
- Fall in mutually exclusive groups
- Finite choice set
- Two types:
- Ordinal
- Categorical

ORDERED MULTINOMIAL DATA

Different flavours
- Proportional odds: same effect on all levels of response
- Generalized ordered models: No relationships proportional
- Partial proportional odds: some relationships are proportional
- Continuation-ratio models: pass through lower values to get to higher

In `R`:
- `polr` function in the `MASS` library
- `vglm` function in the `VGAM` library

CATEGORICAL MULTINOMIAL DATA

Nominal logit model
- Assumes an equation for every outcome
- Probability of each outcome dependent on its own coefficients

Assumption: Independence of irrelevant alternatives
- Preference for one outcome not affected by other possible outcomes
- Strong assumption

Lots of options in `R`: - `multinom` function in `nnet` package
- `vglm` function in `VGAM`
- Other packages that have options include `mlogit`, `geepack`, `MCMCpack` and more

SOME THINGS NEVER CHANGE

Check assumptions
Assess the model
Interpret the model
Stop if things look wonky
A lot of information quickly - You now know options exist - Have the basics to explore